

TIYANA MATTOX

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EDUCATION

Harvard University — Cambridge, MA

Expected May 2028

Bachelor of Science in Mechanical Engineering, Mechanical Systems Track

Secondary in Afro-Latino & Caribbean Studies

Relevant Coursework: Mechanics, MATLAB(EMG signals and Respiratory Analysis), Computational Thinking (Python)

TECHNICAL SKILLS

Programming: MATLAB, Python

CAD & Design: CAD Fusion 360, 3D Printing

Manufacturing: Heat Press Manufacturing, Soldering, Laser Cutting, Prototype Assembly, Jig Design, Polymer Processing, Machine Shop (Mill, Band Saw, Sanders)

Testing: Instron, Mechanical Testing, Peel Testing, Failure Analysis, Design Of Experiments, Data Visualization

ENGINEERING EXPERIENCE

Undergraduate Researcher | Biodesign Lab, Harvard Robotics Core

Jun 2026 – Aug 2026

- Designed and executed Design of Experiments (DOE) investigating dwell time, pressure, temperature, and fixture geometry to optimize manufacturing of flexible biomedical sensors.
- Independently proposed and tested new manufacturing parameters based on experimental results.
- Fabricated 30–40 sensors using heat press manufacturing, laser cutting, and custom assembly fixtures; contributed to production and evaluation of ~50 prototype devices.
- Operated an Instron universal testing system to evaluate peel strength and characterize adhesive versus cohesive failure modes.
- Developed MATLAB scripts to analyze mechanical testing data, evaluate sensor sensitivity, and generate publication-quality figures.
- Designed an improved manufacturing jig in Fusion 360 and prototyped multiple iterations using 3D printing.
- Presented engineering recommendations and research findings to the principal investigator.

ENGINEERING PROJECTS

BreathGuard—Wearable Hardware Enclosure & Sensor Integration: ES 53 Final Project

- Designed a wearable biosensor concept for non-invasive VOC detection, evaluating enclosure architecture, sensor integration, and alternative sensing approaches for a compact wearable device.

Rube Goldberg Machine: Applied Physics 50 Design Project

- Designed and optimized a pulley mechanism to reliably transfer a metal ball between stages, considering load, friction, and energy transfer.
- Built and tested interconnected mechanical mechanisms, iterating on ramps, pulley geometry, and magnetic-car interactions to improve repeatability across trials.

Renal System Simulation—MATLAB: ES 53 Lab

- Developed and analyzed a MATLAB model of renal filtration, simulating how changes in GFR and creatinine production affect plasma concentration and steady-state behavior under physiological conditions.

LEADERSHIP & ACTIVITIES

Social Chair, Harvard Society of Black Scientists & Engineers (NSBSE)

Oct 2025 – Present

- Plan professional, networking, and community-building events for engineering students.

Harvard Varsity Cheerleading — Student-Athlete

June 2024 – Present

- Recipient of the 2025 Unsung Hero Award; balance athletics with engineering coursework.

Community Volunteer

Sep 2025 – Present

- PBHA Y2Y Harvard Square Shelter: Volunteered 12+ hours supporting evening operations for youth experiencing homelessness (8:45pm–11:15pm shifts); PBHA KSNAP: led structured enrichment activities for 15 children with special needs